MAHATMA GANDHI UNIVERSITY KOTTAYAM

FOURTH SEMESTER BCom (Honors)

Model Question Paper

DSC A Quantitative Techniques for Business Decisions. (MG4DSCCOM201)

Time 2 hours **Maximum Marks 70**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Section A

(Answer any 12 questions. Each question carries 1 Mark.)

1. What is Spurious Correlation? (U,CO 1)
2. If the correlation coefficient is 0.9 and bxy = 1.5, find the value of the regression coefficient of Y on X? (A,CO2)
3. Which Correlation method is most suitable for studying the relationship between qualitative data? (U,CO 1)
4. Which statistical measure quantifies the average distance that the observed values fall from the regression line? (U,CO 2)
5. Business recession and recovery indicate which component of time series? (U,CO 3)
6. Which component of time series shows regular repeating changes within a year? (U,CO 3)
7. Why classical approach of probability is also called as “a priori probability? (U,CO 4)
8. An experiment that has two or more outcomes which vary in an unpredictable manner from trial to trial when conducted under uniform conditions is called ----------(U,CO 4)
9. In how many ways can a manager assign a "Cashier" and "Stock Clerk" role to 3 of his employees? (A,CO 4)
10. From 10 different books, how many ways can 3 books be chosen? (A,CO 4)
11. Define Regression Analysis. (U,CO 2)

12. How will you convert annual trend equation into monthly trend equation when the period is even? (A,CO 3)`
13. A customer walks into an electronics store with a budget for only one item. If the probability they buy a smartphone is 0.6 and a smart watch is 0.3, what is the probability they leave the store without buying either item? (A,CO 4)
14. A card is drawn and then another without replacement. Find the probability that both are aces.(A,CO 4)

Section B

(Answer any 4 questions. Each question carries 7 Marks.)

15. Explain the Components of Time Series. (U,CO 3)
16. "Even a high degree of correlation does not mean that a relationship of cause and effect exist between the two correlated variables." Why? (U,CO 1)
17. From the following data, find 4 yearly moving averages.

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	
Sales	25	30	45	50	60	64	75	85	96	(A,CO 3)

18. Out of 6 skilled and 4 unskilled workers, 3 are selected. What is the probability that (a) Two are skilled and one unskilled (b) there is at least one unskilled worker selected(S, CO 4)
19. A store manager has received two boxes of shipment: Box A and Box B, each containing 10 electronic items. Box A contains 3 damaged items and 7 Good items. Box B contains 4 damaged items and 6 Good items. The manager opens one box at random and picks out one item. If the item is found to be damaged, what is the probability that it came from Box A? (An,CO 4)

20. Analyse the direction of price movement over 6 months and find correlation using concurrent deviation. (A, CO 1)

Supply (X)	140	150	155	160	170	180
Price (Y)	80	70	75	65	60	55

Section C

(Answer any 2questions. Each question carries 15 Marks.)

21. Ten contestants are ranked by three judges. Determine which two judges have the most similar taste.(An,CO 1)

Contestant	A	B	C	D	E	F	G	H	I	J
Judge 1	1	2	3	4	5	6	7	8	9	10
Judge 2	4	8	10	2	5	7	6	3	1	9
Judge 3	3	5	2	1	4	10	9	6	7	8

22. The following data relate to the age of the husband and the age of the wife in a small community. (An, CO 2,5)
(a) Find the two regression equations.
(b) Estimate the age of the husband when the age of the wife is 24.
(c) Estimate the age of the wife when the age of the husband is 34.

Age of Husband (X)	25	28	30	32	35	36	38	39	42	45
Age of Wife (Y)	20	26	29	30	25	18	26	35	35	46

23. The annual sales (in ₹ thousands) of OPQ Ltd. are given below:

Year	2019	2020	2021	2022	2023	2024	2025
Sales	150	155	165	170	180	195	205

a) Fit a straight line trend by the method of least squares and tabulate the trend values.

b) Estimate the sales for the year 2026.

c) Eliminate the trend using the additive model.

d) Convert the annual trend equation into a monthly trend equation. (An, CO 3,5)